

From Side Information and Knowledge Graphs to Large Language Models: Two Decades of Knowledge Integration in Recommender Systems

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Abstract

Over the past two decades, Recommender Systems (RSs) have repeatedly changed how they integrate external knowledge, moving from constraints and side information to Knowledge Graphs (KG) and, more recently, Large Language Models (LLMs). This Past/Present/Future retrospective interprets that evolution as a sequence of representational translations rather than simple replacements. Across representative RecSys contributions, we identify recurring patterns: external knowledge is repeatedly used to address sparsity and cold start, human-interpretable structure remains central to explanation and control, and hybrid architectures re-emerge whenever scalability conflicts with traceability. We argue that LLMs should be understood not as substitutes for structured knowledge, but as a new interaction layer that requires explicit grounding, and we conclude with a cautious agenda for hybrid, inspectable, and accountable RSs.

Keywords

recommender systems, side information, knowledge graphs, large language models

1 Introduction

Recommender Systems (RSs) can be viewed as *knowledge systems*: they infer preferences from observations, combine those signals with information about users and items, and present decisions to humans. Over two decades, the RecSys community has continually changed the place knowledge resides and how it is utilized. Early knowledge-based recommenders relied on explicit constraints, domain rules, and critique/repair loops [13, 38]; later work translated knowledge into side information, sparse feature matrices, and latent representations [30, 35, 39]; the following phase reintroduced explicit relational structure through linked open data and knowledge graphs [8, 28, 31]; and the most recent wave treats natural language, processed by large language models (LLMs), as a substrate for preference elicitation, reasoning, and explanation [15, 24, 26]. These phases also diversified internally. The graph phase evolved from semantic-path features to entity-centric representations and interactive interfaces for exploration and explanation [16, 32]. Concurrently, advances in large language models (LLMs) span multiple research and application domains, including alignment and fine-tuning methodologies, sequential recommendation systems, narrative-driven recommendation frameworks, and retrieval-augmented conversational pipelines.[6, 17, 29, 51].

Seen from this perspective, the history of knowledge integration in recommendation is not a sequence of simple replacements, but a sequence of *representational translations*. Each translation makes some capabilities easier to scale and generalize, while making others harder to inspect, communicate, or control. This is why the present

moment deserves a retrospective view: LLM-based recommenders are often presented as a break with the past, yet many of their promises and tensions echo older RecSys questions about sparsity, explanation, traceability, and user control.

This Past/Present/Future retrospective revisits that evolution through three research questions:

- **RQ1 (representational translations)** asks how external knowledge shifted from constraints and rules to features, graphs, embeddings, and language, and what each translation gained or obscured;
- **RQ2 (recurring design patterns)** asks which patterns repeatedly re-emerge across eras, such as using richer side information to fight sparsity, introducing human-meaningful structure to make reasons communicable, and separating heavy offline enrichment from lightweight online decision-making;
- **RQ3 (responsible knowledge integration)** asks what new failure modes arise when knowledge integration becomes generative and conversational, and how they relate to the older knowledge-based promise of traceability.

We argue that new approaches do not *replace* structured knowledge such as graphs, attributes, and constraints. Rather, they create a new, highly expressive *interaction layer*. To remain trustworthy and deployable, that layer must be grounded in explicit knowledge stores and evaluated with protocols that are reproducible, fairness-aware, and sensitive to generative failure modes.

Contributions. This paper offers three contributions: a compact lens of *knowledge integration as representational translation*, a synthesis of recurring design patterns across algorithmic families, and a cautious 2030 agenda for hybrid, grounded, and accountable RSs.

2 Corpus and coding methodology

This retrospective analyzes the structured reading of RecSys proceedings and closely related literature from the same research trajectory that explicitly examines the integration of external knowledge into recommendation systems. We use the term *external knowledge* in a broad yet practical sense, which encompasses item and user metadata, declarative rules and constraints, semantic and linked data, knowledge graphs, natural language resources, retrieval components, and studies focused on explanation or reproducibility, as long as they contribute to the knowledge integration process. To make the review process more transparent and reusable, we provide a companion repository¹.

We exclude studies that employ side information solely as a modeling convenience; to be admitted to the corpus, a work must render the knowledge layer itself explicit within at least one component

¹Companion repository: https://split.to/KG_PPF

of the modeling, explanatory, interaction, or evaluation pipeline². Our goal is not to provide an exhaustive bibliographic coverage. Instead, consistent with the Past/Present/Future format, we focus on representative papers that make one of three roles visible: a representational shift, a recurring design pattern, or an emerging failure mode. This offers a clear discussion for comparing various RS families.

Coding scheme. To compare contributions written in different eras and under different evaluation traditions, we code each paper along five dimensions: (1) *knowledge representation* (constraints/rules; feature matrices; structured graphs; learned embeddings; natural language/LLM); (2) *knowledge role* (prediction; cold-start/sparsity mitigation; explanation/control; interaction; evaluation); (3) *integration locus* (offline enrichment; online inference; hybrid); (4) *evaluation style* (offline ranking; user study; system/reproducibility study; qualitative/position); (5) *risk surface* (fairness; hallucination/out-of-domain (OOD) errors; reproducibility; governance/accountability). These dimensions are intentionally lightweight: they are not used to rank papers, but to support the cross-era comparisons behind RQ1–RQ3 and the resulting 2026–2030 agenda.

Threats to validity. This retrospective necessarily privileges interpretability over completeness. Selection effects remain possible, representative papers may understate parallel developments, and some category boundaries are porous (i.e., graphs can support both explicit reasoning and latent models, while LLMs systems often combine retrieval, ranking, and generation). In addition, some relevant innovations emerged outside core RecSys venues. We mitigate risks by using familiar RecSys examples and external works to clarify transitions. Our coding dimensions are explicit to ensure transparency in interpretive choices.

3 Knowledge representations as recurring translations

RQ1: How has the representation of external knowledge shifted over two decades, from constraints and rules, to feature matrices, to structured graphs, to learned embeddings, to natural language, and what capabilities were gained and lost?

Rather than reading this history as a linear march toward “better” models, we interpret it as a sequence of *representational translations*. Each step increases a combination of scale, coverage, or interaction bandwidth, while making it more difficult to maintain some level of traceability, explicit semantics, or direct user control.

From constraints and rules to learnable representations. Early knowledge-based RSs represent knowledge declaratively: recommendations are derived by satisfying constraints, applying rules, and iteratively repairing failing queries. This yields high *communicability* (the system can explain which constraints were violated and propose relaxations) and high *controllability* (users can manipulate constraints or critiques), but it creates a knowledge acquisition and maintenance bottleneck and limits scalability [13, 25, 38].

Intermediate work on knowledge infusion has already foreshadowed the next translation by moving domain knowledge into richer item descriptions and feature spaces rather than leaving it only

in the reasoning procedure [39]. The subsequent shift to sparse feature matrices (via linear models, matrix factorization variants, and “everything is a feature” approaches) improves scalability and benchmarking maturity, and makes side information “pluggable” as covariates in standard learning pipelines [30, 35, 41]. The cost is that relational semantics are absorbed into weights and embeddings, and explanations often become post-hoc reconstructions rather than direct traces of the decision process. Research on side information shows that this translation remained relevant beyond the initial FM/SLIM wave, using variational and closed-form approaches that treated metadata as key input for collaborative models. [11, 19].

From semantic graphs to embeddings and language. The linked open data (LOD) and knowledge-graph (KG) phase reintroduces explicit relational structure at scale. LOD-based methods extract semantic paths and features from knowledge bases such as DBpedia to improve recommendation under sparsity [31], while explainability-oriented pipelines make that structure visible to users by verbalizing properties and relations as recommendation rationales [28]. Logic- and graph-based systems further show that knowledge graphs are especially useful in sparse or cold-start settings, and often deliver the most value when combined with latent factorization rather than treated as a standalone replacement [8]. The same idea also diversified into entity-centric learning, KG-aware autoencoding, sparse feature factorization, and neuro-symbolic combinations of graph embeddings with logical rules [3, 7, 32, 40]. Structured knowledge is not a single technique but a collection of methods that clarify semantics while preserving predictive power.

A further translation compresses metadata and graph structure back into learned vectors. Metadata-aware embedding models such as Meta-Prod2Vec inject side information during training, while KG embedding approaches reduce manual path engineering and support scalable downstream inference [42, 44]. What this step gains in portability, efficiency, and generalization, it partly loses in the immediate visibility of why a recommendation was produced.

LLM-centric RSs push the translation one step further by treating interactions, metadata, and reviews as language sequences and by framing recommendation tasks in text-to-text terms [15]. This move greatly expands interaction bandwidth for conversational preference elicitation, explanation synthesis, and interface generation [26]. At the same time, it makes provenance, constraint satisfaction, and auditability more fragile unless the model is explicitly grounded in retrieval pipelines or external knowledge stores [22, 24]. Within the RSs landscape, that language-centered translation already appears in several forms: alignment/tuning methods [6], metadata and item-description generation [1, 2], open-world augmentation [50], and industrial reasoning layers that distill or retrieve LLM-produced rationales into deployable RSs [43].

Across these translations, external knowledge never disappears; it migrates between layers. Sometimes it is carried by rules, sometimes by features, sometimes by entities and relations, and increasingly by prompts, retrieved evidence, or interface state. This persistent migration helps explain why hybrid architectures repeatedly re-emerge across eras.

²We sample tutorials, workshops, reproducibility studies, and industrial reports from the recent LLM phase, as this area is influenced by new models, evaluation frameworks, deployment reports, and tools.

4 Recurring design patterns across eras

RQ2: What design patterns reappear across eras, and how do they manifest in distinct algorithm families?

The historical sequence in Section 3 illustrates consistent responses to ongoing challenges in RecSys. Across different periods, the field has repeatedly utilized similar design strategies: enhancing sparse signals with external knowledge, maintaining understandable structures for explanations, processing complex knowledge offline, and combining complementary components.

Pattern 1. Fight sparsity with external knowledge. This pattern is perhaps the most stable of all. Early knowledge-based RSs already reduce their dependence on dense behavioral logs by using domain knowledge and explicit requirements [13]. Side information models make the same move statistically scalable by injecting metadata into feature spaces for top- N recommendation under sparse interactions [30]. LOD/KG approaches extend it again by using semantic paths and entity relations to recover useful signal in sparse or cold-start settings [8, 31]. In the LLM era, language-based recommenders can remain competitive in near cold-start settings when user preferences are expressible in text and can complement item-based collaborative filtering [37]. Across all four cases, external knowledge acts as a surrogate signal whenever interaction evidence is too weak to stand alone.

Recent industrial and application papers show the same logic in operational form: LLMs are used to generate richer item descriptors, support story discovery, or widen accessible knowledge when interaction evidence is thin [18, 23, 50].

Pattern 2. Make reasons communicable by introducing a human-understandable structure. A second recurring theme is the effort to ensure that the reasons for the recommendation are understandable to humans. In constraint-based systems, the clarity of communication is inherent in the representation itself: the system can reveal which requirements failed and what can be adjusted [38]. LOD/KG-based explanation frameworks scale that intuition by translating entity-relation structure into natural-language rationales and measuring their user impact [28]. LLM-era Conversational RecSys (CRSs) further expand communicability through generated language and UI elements, but they also show that fluency is not enough: without retrieval support or structured grounding, reasons may sound persuasive while remaining weakly tied to the actual evidence [26]. The repeated lesson is that human-meaningful structure is not only a UX asset; it is also a mechanism for aligning recommendations, explanations, and user control [16, 33, 49].

Pattern 3. Separate offline enrichment from lightweight online decision-making. A third pattern concerns the division of labor between heavy knowledge processing and fast-time interaction. Meta-Prod2Vec, for instance, injects metadata during training so that online recommendation can operate over compact embeddings [44]. KG-enhanced Neural Networks (KGNN) recommenders make the same point from a systems angle: richer relational reasoning is useful, but training and serving require memory-aware optimizations to remain deployable [9]. In the LLM era, the tension becomes even sharper because generation is expressive but costly, motivating teacher-student distillation and hybrid retrieval+generation pipelines [10]. Although the implementation changes across eras,

the architecture principle that expensive enrichment should not dominate the latency-critical path stays constant. Industrial reports make this architecture principle explicit: LLMs often expand the interest space, annotate content, or provide teacher signals that are later cached, retrieved, or distilled into faster serving-time components [36, 43, 46].

Pattern 4. Hybridize rather than replace. Finally, new knowledge representations rarely cleanly eliminate older ones. Approaches explicitly combining graphs with latent factorization instead of choosing one over the other have been proposed [8]. RAG later formalizes the same hybrid logic as a combination of parametric and non-parametric memory, making provenance and updatability first-class concerns [22]. Recommender-specific grounding work continues this pattern by bridging retrieval and generation to avoid OOD (Out Of Distribution) or hallucinated items [54]. Hybridization reappears as each layer offers unique contributions such as coverage, semantics, controllability, or interaction fluency. The best systems arise from composition rather than replacement. Recent hybrid RSs continue this logic by aligning collaborative and semantic representations, revisiting KG representation learning with generative models, or balancing fine-tuning with retrieval for dynamic updates [5, 27, 47].

Taken together, these recurring patterns suggest that the field’s core problems are more stable than its model families. This continuity is also why the LLM transition should be read with caution: it changes the surface of interaction, but it does not remove the need for grounding, inspectability, and explicit evaluation contracts.

5 New failure modes and responsible knowledge integration

RQ3: What new risks emerge when knowledge integration becomes generative and conversational, and how do they relate to the older knowledge-based promise of traceability?

These risks are not entirely separate from previous work in KG RSs; they reflect *old* concerns, like validity, fairness, accountability, and user protection, that are reappearing in a new interaction context. When recommendations are generated as open-ended language, the system not only ranks items but also decides what to mention, how to justify its choices, and how confidently to present them. Failures can now appear at the interface level through generated items and justifications, before the user sees a ranked list. While this enhances expressiveness, it also increases the risk of errors.

Fairness becomes output- and prompt-sensitive. Classic fairness metrics assume access to calibrated scores on a fixed candidate set; LLM-based recommendation can violate both assumptions. FaiRLLM therefore introduces a benchmark with tailored metrics and a dataset spanning multiple sensitive attributes, showing that ChatGPT can still exhibit unfairness in generated recommendations [52]. The implication is not that ranking-based fairness notions disappear, but that responsible evaluation must additionally target *generated outputs* and their dependence on the generative interface. **Hallucinations and out-of-domain items are decision-support failures.** In earlier knowledge-based systems, invalid recommendations were often observable as violated constraints or failed repairs [13, 38]. In LLM-based Conversational RSs (CRSs), by contrast, failures may surface as fluent but unsupported suggestions. Recent

work on graph retrieval-augmented CRSs notes that hallucinated recommendations remain a key issue when domain knowledge is incomplete [34], while work on OOD recommendation treats unsupported items as a central failure mode and proposes grounding/constrained generation to eliminate them [54]. This reframes grounding as more than a Natural Language Generation quality improvement: it is a core requirement for reliable recommendation. Related work on catalog alignment and open-world recommendation makes the same point from the systems side: generative flexibility is only useful when the model remains anchored to a valid, current item universe [45, 50].

Reproducibility and accountability require explicit tooling. A reproducibility study of the Personalized Prompt & Predict (namely, P5) paradigm argues that LLM-based recommenders can help address cold-start and sparsity issues, but also introduce accountability challenges and benchmarking uncertainty [24]. Tools such as LaikaLLM make this concern operational by supporting more reproducible training and evaluation pipelines³. In parallel, LLM-as-evaluator work suggests a scalable approach for assessing recommendation explanations via meta-evaluation against user feedback, while also demonstrating sensitivity to prompts and settings [53]. These results suggest that prompts, retrieval corpora, decoding choices, and evaluation instructions are essential to the experimental method, not just hidden details.

Privacy and security enter the knowledge-integration risk surface. Because LLM-based RSs can encode detailed textual histories and preference profiles, the knowledge layer itself can become an attack surface. Recent studies highlight inversion attacks on LLM-powered RSs and textual attacks on news recommendations, emphasizing that grounding, safety, and privacy protection should be treated as interconnected design constraints. [48, 55].

Across these failure modes, the underlying requirement is consistent with older knowledge-based aspirations: a recommender should be able to justify what it recommends, remain inspectable when it fails, and expose enough evidence for users and researchers to contest or reproduce its behavior. In the generative setting, this need becomes measurable through fairness benchmarks, hallucination/OOD checks, grounding compliance, and reproducible evaluation pipelines. This motivates the agenda in the next section.

6 A 2026–2030 agenda for knowledge integration

The retrospective above suggests four design priorities rather than a single perspective.

Grounded generation with inspectable provenance. Generative RSs should treat grounding as one of their objectives. Recommended items and attached rationales should be backed by explicit evidence in a retrievable store (e.g., catalog metadata, reviews, logs, or a KG) so that outputs remain updatable and auditable. This aligns with RAG’s emphasis on provenance and with RecSys prototypes that use retrieval augmentation to reduce unsupported claims [22, 26, 54]. Recent work on catalog alignment and dynamic update mechanisms suggests that this grounding requirement also includes keeping the item universe synchronized with changing catalogs and user interests over time [27, 45].

³<https://github.com/swapUniba/LaikaLLM>

Hybrid retrieval+ranking+LLM stacks under deployment constraints. Future systems should be designed explicitly as hybrids: a retrieval/ranking core paired with an LLM layer for dialogue, critique handling, explanation, or interface adaptation, or powerful LLMs used as teachers for smaller deployable students [10]. This is consistent with the recurring separation between heavy offline enrichment and lightweight online decision-making, and with systems work showing that infrastructure constraints remain decisive for deployment [9, 21]. Industrial deployments and distillation-based approaches already point in this direction, using LLMs to expand interest spaces or to populate controllable reasoning pools while preserving conventional serving architectures [36, 43, 46].

Evaluation as a full pipeline specification. For recommendations with LLMs work to accumulate, prompts, candidate constraints, retrieval corpora, decoding settings, and randomness control must be treated as part of the method. Reproducibility studies already show that these choices affect conclusions [24], and toolkits such as LaikaLLM³ show how more controlled experimentation can be operationalized. Shared reporting checklists would, for generative recommenders, play a role similar to standardized offline protocols in earlier RecSys eras. At minimum, LLM-based papers should report whether recommendations are generated freely or selected from a fixed candidate slate, because that choice changes both failure modes and the meaning of offline metrics. Evaluation may also need stronger meta-evaluation tools, but only under explicit reporting of their prompts, rubrics, and failure cases [12, 14, 20].

Responsible integration as a measurable contract. Responsible knowledge integration should be evaluated, not merely asserted. Papers should report fairness and safety outcomes alongside utility, and explicitly measure hallucination/OOD rates and grounding compliance [4, 52, 54]. More broadly, systems should expose enough evidence for users and researchers to inspect failures, extending the older knowledge-based aspiration of traceability into a generative setting. That contract should arguably include privacy and robustness audits as well, given the growing evidence that LLM-based RSs can leak user signals or be steered by adversarial text [48, 55].

These priorities describe conditions under which richer interaction can remain compatible with control, inspectability, and scientific comparability. Under this lens, future RecSys papers should state where knowledge lives, how it is grounded, and what parts of the pipeline remain inspectable.

7 Conclusion

This retrospective argues that RecSys has not moved from the “knowledge-rich” to the “knowledge-free” paradigms, but from one way of encoding knowledge to another. Constraints and critiques, feature matrices, graphs, embeddings, and language each redistribute what can be scaled, explained, or controlled. Read in this way, the LLM turn is neither a rupture nor an endpoint. It is the latest representational translation, one that expands interaction and reasoning bandwidth while making grounding and accountability more urgent. The practical implication is clear: progress should favor hybrid architectures that keep explicit knowledge available for provenance, user control, and evaluation, and should judge success by groundedness, reproducibility, and responsible behavior alongside utility.

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